

Macroeconomics I (ECO387C) Midterm Exam (Mock)

Fall 2026

Name: _____

Instructions. Closed book, no notes. Answer all four questions. Show your derivations, since most of the credit is for the steps. Where a number is asked for, leave it as a fraction or simple expression if you have no calculator. Total 100 points, about 2 hours. Write your answers in the space below each part and continue on the back if needed.

Question 1. A stochastic endowment economy (25 points)

A continuum of identical households maximizes $E_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma}$, $0 < \beta < 1$, $\gamma > 0$. Households trade a riskless one-period bond. Buying a_{t+1} units at t pays $R_t a_{t+1}$ units of goods at $t + 1$, where R_t is the gross rate set at t . Bonds are in zero net supply. Every household receives the same endowment $y_t \in \{y_h, y_l\}$ with $y_h > y_l > 0$. The endowment follows a Markov chain. From y_h it stays at y_h with probability p_h , and from y_l it stays at y_l with probability p_l .

(a) (2 points) Write the household's flow budget constraint.

(b) (3 points) What are the state variables of the household problem? Explain why the current endowment is a state even though it is exogenous.

(c) (3 points) Write the Bellman equation, with the expectation written out as a sum over next period's states.

(d) (5 points) Derive the Euler equation from the Bellman equation. Label the first-order condition and the envelope condition.

(e) (4 points) State the market-clearing conditions. Use them to show that in equilibrium $a_{t+1} = 0$ and $c_t = y_t$, and give a formula for the equilibrium gross rate $R(y)$.

(f) (4 points) Let $\gamma = 2$, $y_h = 2$, $y_l = 1$. Find R_h and R_l as functions of β , p_h , p_l . Evaluate them at $\beta = 0.95$, $p_h = 0.8$, $p_l = 0.6$ and compare with $1/\beta$.

(g) (4 points) With the same numbers, the log case ($\gamma = 1$) gives $R_h = 0.877$ and $R_l = 1.316$. Explain why the rates are further from $1/\beta$ when $\gamma = 2$. Then suppose the endowment is iid, $p_h = 1 - p_l$. Show that R_l/R_h no longer depends on the probabilities and find it.

Question 2. An anticipated income rise (25 points)

Time is $t = 0, 1, 2, \dots$. There are two household types, A and B , each of mass one, with preferences $\sum_{t=0}^{\infty} \beta^t \ln c_t^e$ for $e \in \{A, B\}$. Type A receives $y_t^A = 1$ every period. Type B receives $y_0^B = 1$ and $y_t^B = 2$ for all $t \geq 1$. Everybody knows this at $t = 0$. Households trade one-period real bonds: $c_t^e + b_t^e = y_t^e + (1 + r_{t-1})b_{t-1}^e$ with $b_{-1}^e = 0$.

(a) (4 points) State the household problem, including the no-Ponzi-game condition, and define a sequential competitive equilibrium.

(b) (4 points) Derive the Euler equation and the intertemporal budget constraint. In the intertemporal budget constraint, write $q_t := 1 / \prod_{\tau=0}^{t-1} (1 + r_{\tau})$.

(c) (4 points) Show that $1 + r_t = \frac{1}{\beta} \frac{Y_{t+1}}{Y_t}$ with $Y_t := y_t^A + y_t^B$. Find r_0 and r_t for $t \geq 1$, and the prices q_t . Explain why r_0 is high.

(d) (6 points) Solve for c_t^A and c_t^B for all t . Verify that the goods market clears.

(e) (4 points) Find b_0^A and b_t^A for $t \geq 1$. Who borrows at $t = 0$, and why? Check that A 's bond path satisfies the no-Ponzi-game condition.

(f) (3 points) In the Arrow-Debreu version of this economy (textbook Section 5.2.1), the price of date- t goods in units of date-0 goods is $p_t = \beta^t Y_0 / Y_t$. Show that $p_t = q_t$ and explain in one sentence why the two market structures give the same allocation.

Question 3. Growth by dynamic programming (25 points)

A planner maximizes $\sum_{t=0}^{\infty} \beta^t \ln c_t$ subject to $c_t + k_{t+1} = Ak_t^\alpha$, $k_{t+1} \geq 0$, $k_0 > 0$ given, with $0 < \alpha < 1$, $0 < \beta < 1$, $A > 0$ (full depreciation).

(a) (3 points) Write the problem in the form $\max \sum \beta^t F(k_t, k_{t+1})$ with $k_{t+1} \in \Gamma(k_t)$. Give F and Γ , and write the Bellman equation.

(b) (5 points) Let T be the Bellman operator. State Blackwell's sufficient conditions and verify both for T . What does Banach's fixed-point theorem then deliver? Why is some care needed with log utility?

(c) (5 points) Start value function iteration from $V_0 \equiv 0$. Find V_1 and its policy. Then find the policy that maximizes the right-hand side when computing V_2 , and interpret it as a decision in a finite-horizon problem.

(d) (6 points) Derive the functional Euler equation using the first-order and envelope conditions. Guess $g(k) = sAk^\alpha$ and solve for s .

(e) (3 points) Find the steady state k^* and evaluate it at $\alpha = 0.3$, $\beta = 0.95$, $A = 1$.

(f) (3 points) Write the policy in log deviations $\hat{k}_t := \ln k_t - \ln k^*$. If $k_0 = k^*/2$, what is $\hat{k}_1$? How many periods does it take to close half of the gap?

Question 4. Consumption (25 points)

A household maximizes $E_0 \sum_{t=0}^{\infty} \beta^t u(c_t)$ subject to $a_{t+1} = (1+r)(a_t + y_t - c_t)$ and a no-Ponzi-game condition, with $\beta(1+r) = 1$ and $r > 0$. Income y_t is stochastic.

(a) (3 points) Let $u(c) = b_1 c - \frac{1}{2} b_2 c^2$. Show that $c_t = E_t c_{t+1}$.

(b) (5 points) Iterate the budget constraint forward and use part (a) to show

$$c_t = \frac{r}{1+r} \left[a_t + \sum_{j=0}^{\infty} \left(\frac{1}{1+r} \right)^j E_t y_{t+j} \right].$$

(c) (6 points) Let $y_t - \bar{y} = \rho(y_{t-1} - \bar{y}) + \varepsilon_t$ with $0 \leq \rho \leq 1$ and ε_t white noise. Show that $\Delta c_t = \frac{r}{1+r-\rho} \varepsilon_t$. Evaluate the coefficient at $r = 0.04$ for $\rho = 0, 0.9, 1$ and interpret.

(d) (3 points) Hall (1978) regresses Δc_{t+1} (or $\Delta \ln c_{t+1}$) on variables known at t . What does the model predict for the coefficient and why? What did Johnson, Parker and Souleles (2006) find?

(e) (5 points) Now add the borrowing constraint $a_{t+1} \geq 0$ and let income be iid with mean $\bar{y} = 1$, so $\rho = 0$. At $a_t = 0$ the household draws $y_t = 0.5$, with $r = 0.04$. Compute the consumption the unconstrained rule from (b) prescribes, and show that it violates the constraint. Write the Euler equation with the constraint's multiplier, and give consumption and the marginal propensity to consume at this point.

(f) (3 points) Replace quadratic utility by CRRA. Explain, using a second-order expansion or a picture, why the household now saves more when income risk rises, even with no borrowing constraint.